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Roll Number : 31/A		Lab Practical Number: 10	
Title of LAB Practical : To develop Application for Windows Azure / Amazon AWS using Windows Azure Platform Training Kit and Visual Studio.			
DOP : 02/10/2025		DOS : 12/10/2025	
CO Mapped :	PO Mapped:	Faculty Signature:	Marks :

Title: To develop Application for Windows Azure / Amazon AWS using Windows Azure Platform Training Kit and Visual Studio.

Aim : To develop Application for Windows Azure / Amazon AWS using Windows Azure Platform Training Kit and Visual Studio.

Theory :

Cloud computing provides a platform for delivering computing services such as servers, storage, databases, and software over the Internet, allowing applications to be deployed without managing physical infrastructure. Microsoft Azure is a cloud platform offering Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). It enables developers to build, test, and deploy applications quickly and efficiently.

Using **Visual Studio** with the **Azure SDK**, developers can seamlessly integrate their local projects with cloud services. ASP.NET Core MVC allows the creation of dynamic web applications with clear separation between the presentation, logic, and data layers. Through Azure App Service, applications can be deployed and scaled automatically while maintaining high availability.

The **Windows Azure Platform Training Kit** provides resources and tools that simplify the learning and development process by offering templates, sample projects, and tutorials. Azure also supports automatic configuration, HTTPS hosting, monitoring, and global content delivery. By connecting Visual Studio to Azure, developers can perform end-to-end development—from code creation to live deployment—within a single environment.

Tools and Technologies Used:

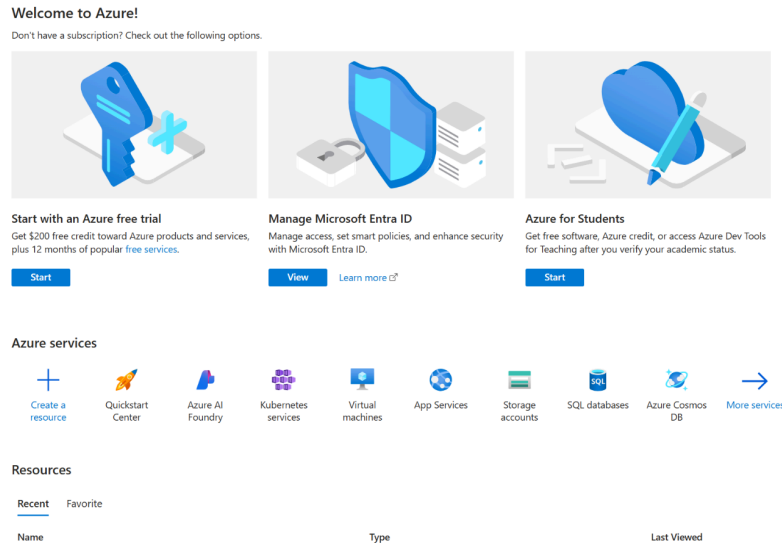
- **Visual Studio (2019 or later)**
- **Windows Azure Platform Training Kit**
- **Microsoft Azure Account / Amazon AWS Account**
- **.NET Framework / .NET Core**
- **Azure SDK for .NET / AWS SDK for .NET**
- **SQL Azure / AWS RDS (for database)**

STEP 1: Setup Environment

1. Install Visual Studio with Azure development workload:

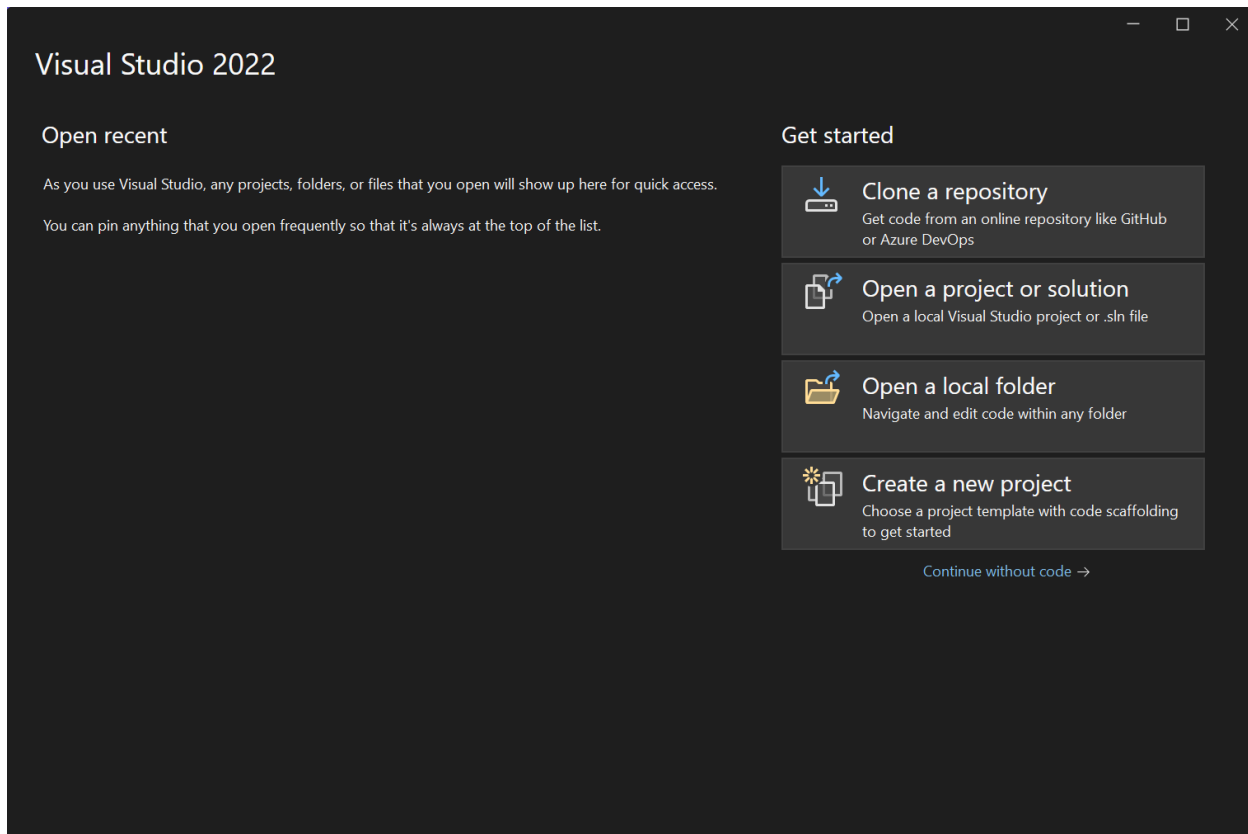
- Open Visual Studio Installer → Modify → Check “Azure Development” → Install.

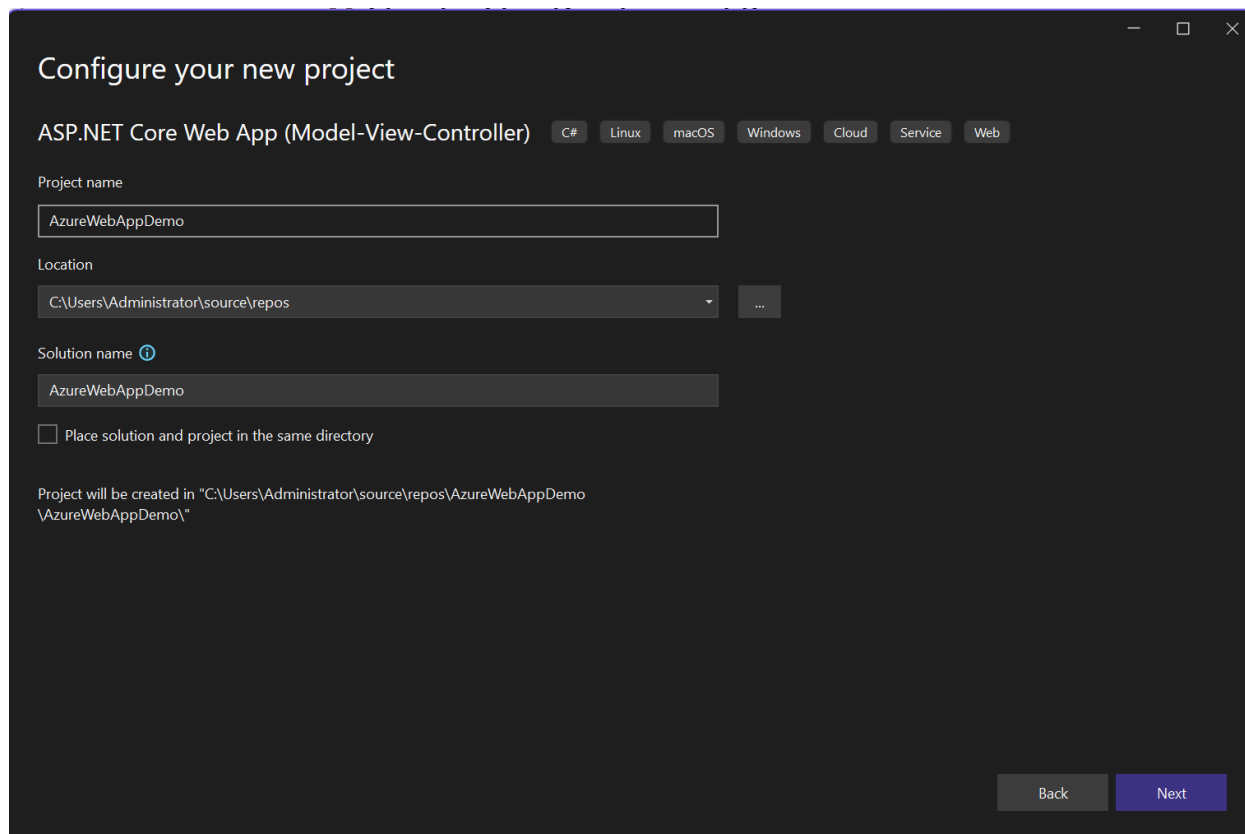
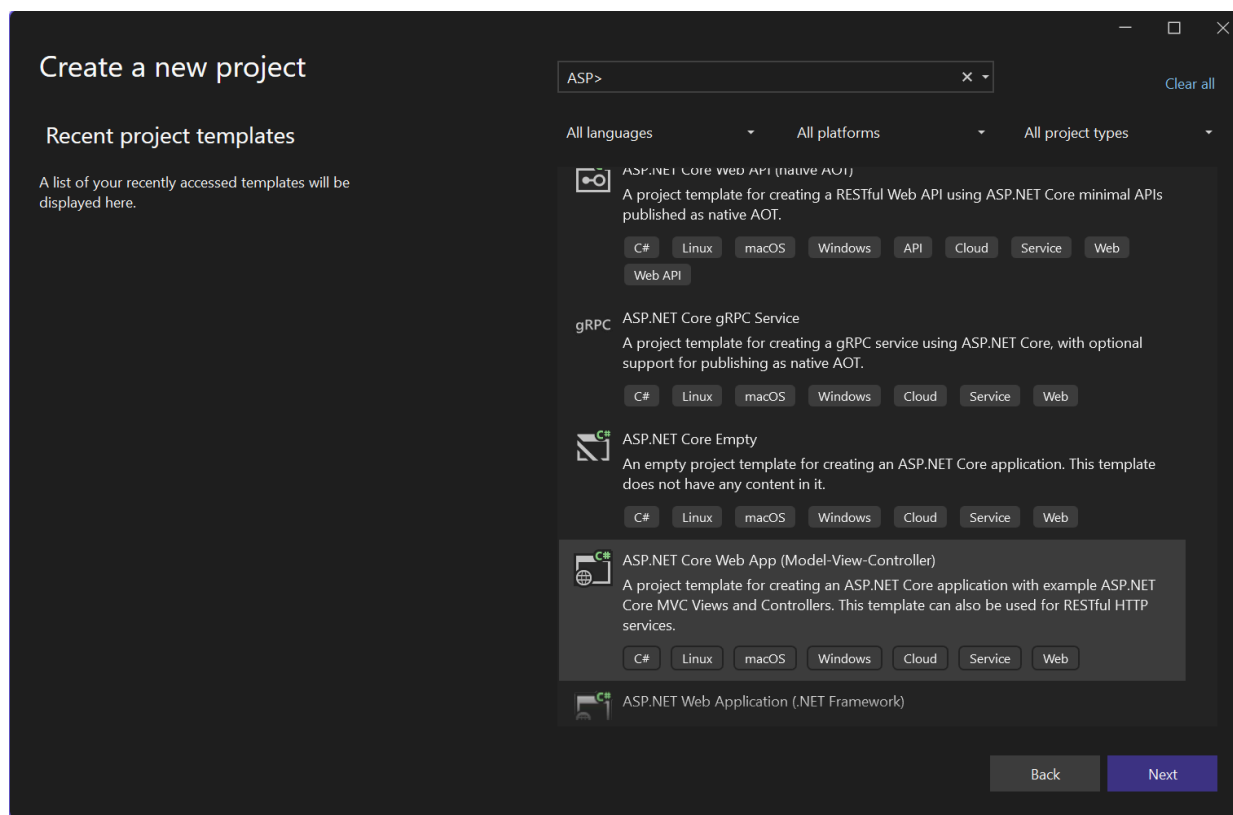
2. Ensure you are signed in to **Visual Studio** with your **Microsoft account** (same as your Azure account).
3. Create a **free Azure account** if you don't have one at <https://portal.azure.com>



STEP 2: Create a New Project

1. Open Visual Studio → **Create a new project**
2. Select **ASP.NET Core Web App (Model-View-Controller)** → Next
3. Project Name: AzureWebAppDemo
4. Framework: **.NET 8 (Long-term support)** → Create





Additional information

ASP.NET Core Web App (Model-View-Controller) C# Linux macOS Windows Cloud Service Web

Framework ⓘ
[.NET 8.0 (Long Term Support)]

Authentication type ⓘ
[None]

☒ Configure for HTTPS ⓘ
☐ Enable container support ⓘ

Container OS ⓘ
[Linux]

Container build type ⓘ
[Dockerfile]

☐ Do not use top-level statements ⓘ
☐ Enlist in .NET Aspire orchestration ⓘ

Aspire version ⓘ
[9.2]

Back Create

STEP 3: Write Application Code

Below is a minimal MVC example.

Code:

Controller/[HomeController.cs](#)

```
using Microsoft.AspNetCore.Mvc;

namespace AzureWebAppDemo.Controllers
{
    public class HomeController : Controller
    {
        // GET: Home/Index
        [HttpGet]
        public IActionResult Index()
        {
            return View();
        }
    }
}
```

```
// POST: Home/Index
[HttpPost]
public IActionResult Index(string userName)
{
    if (string.IsNullOrEmpty(userName))
    {
        ViewBag.Message = "Please enter your name.";
    }
    else
    {
        ViewBag.Message = $"Hello {userName}, welcome to Azure Cloud!";
    }

    return View();
}

// Optional Privacy page (you can remove if not needed)
public IActionResult Privacy()
{
    return View();
}
}
```

Views/Home/Index.cshtml

```
@{
    ViewData["Title"] = "Azure Web App Demo";
}

<div style="text-align:center; margin-top:80px; font-family:Segoe UI, sans-serif;">
    <h1>@ViewData["Title"]</h1>
    <h3>Enter your name to see Azure in action!</h3>

    <form method="post" style="margin-top:20px;">
        <input type="text" name="userName" placeholder="Enter your name"
            style="padding:8px; width:250px; border-radius:5px; border:1px solid #ccc;"
        required />
        <button type="submit"
            style="padding:8px 16px; background-color:#0078D7; color:white; border:none;
            border-radius:5px;">
            Submit
        </button>
    </form>

    @if (ViewBag.Message != null)
    {
```

```
        <h3 style="color:green; margin-top:30px;">@ViewBag.Message</h3>
    }
</div>
```

Program.cs

```
var builder = WebApplication.CreateBuilder(args);

// Add services to the container.
builder.Services.AddControllersWithViews();

var app = builder.Build();

// Configure the HTTP request pipeline.
if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    app.UseHsts();
}

app.UseHttpsRedirection();
app.UseStaticFiles();

app.UseRouting();

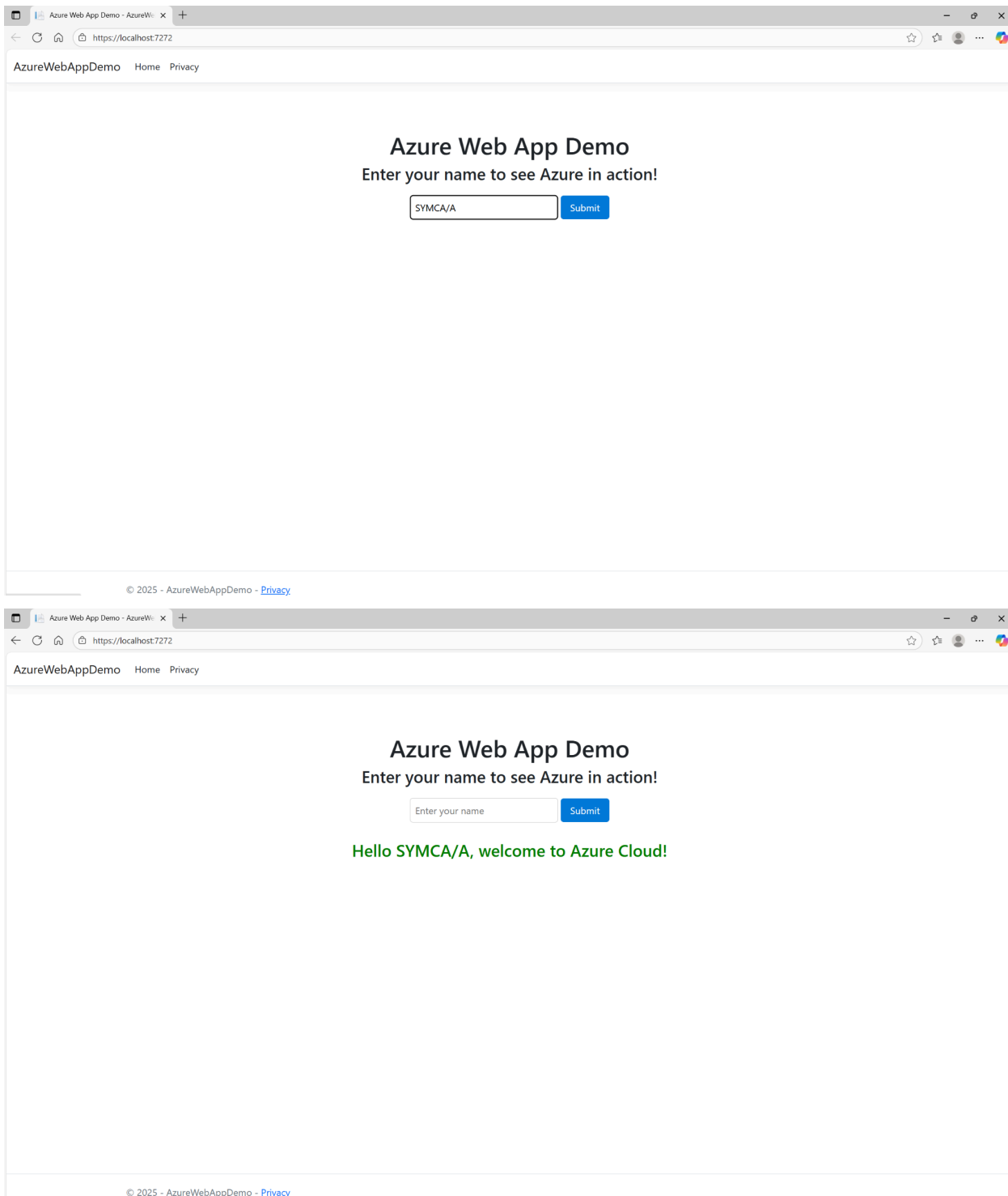
app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
    pattern: "{controller=Home}/{action=Index}/{id?}");

app.Run();
```

STEP 4: Run Locally

1. Press Ctrl + F5 → The app opens in your browser at <https://localhost:xxxx/>.
2. Enter your name → Check that it displays “Hello [Name], welcome to Azure Cloud!”

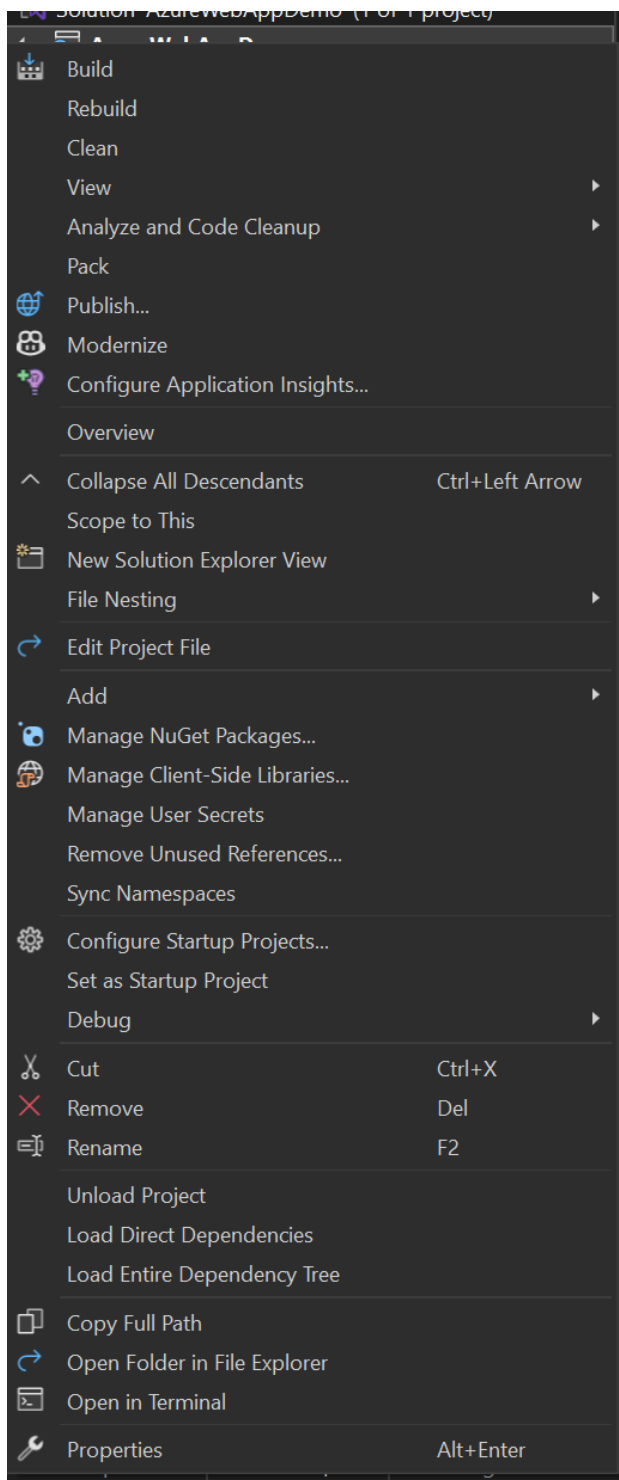


Step 5 – Deploy to Azure

Once you confirm it's working locally:

1. Right-click your project → Publish
2. Select Azure → Azure App Service (Windows) → Next

3. Sign in with your Azure account
4. Create a new App Service (Free F1 tier)
5. Click Publish



Publish

Where are you publishing today?

Target

**Azure**
Host your application to the Microsoft cloud

**Docker Container Registry**
Publish your application to any supported Container Registry that works with Docker images

**Folder**
Publish your application to a local folder or file share

**FTP/FTPS Server**
Publish your application to an FTP/FTPS server

**Web Server (IIS)**
Publish your application to IIS using Web Deploy or Web Deploy Package

**Import Profile**
Import your publish settings to deploy your app

Back

Next

Finish

Cancel

Publish

Which Azure service would you like to use to host your application?

Target

Specific target

**Azure Container Apps (Linux)**
Run scalable containerized applications and microservices on a serverless platform in Azure

**Azure App Service (Windows)**
Publish your application code to a managed infrastructure that is easy to scale

**Azure App Service (Linux)**
Publish your application code to a managed infrastructure that is easy to scale

**Azure App Service Container**
Publish your application as a Docker image to Azure Container Registry and run it on Azure App Service

**Azure Container Registry**
Publish your application as a Docker image to Azure Container Registry

**Azure Virtual Machine**
Manage your own infrastructure

Back

Next

Finish

Cancel

Select existing or create a new Azure App Service

Target

Subscription name

Azure for Students

Specific target

App Service

Search

+ Create new [Filter] [Refresh]

There are no existing instances available

Create a new instance

☐ Deploy as ZIP package

☐ Turn on Basic Authentication (not recommended) ⓘ

Back Next Finish Cancel

Name

azurewebappdemo123

Subscription name

Azure for Students

Resource group

AzureWebAppDemo20251010224951ResourceGroup*

New...

Hosting Plan

AzureWebAppDemo20251010224951Plan* (Canada Central, S1)

New...

☒ Use unique default hostname (preview).

Export... Create Cancel

Name
azurewebappdemo123

Subscription name
Azure for Students

Resource group
AzureWebAppDemoRG* New...

Hosting Plan
AzureWebAppDemo20251010224951Plan* (South India, F1) New...

☒ Use unique default hostname (preview).

 Creating: App Service... Create Cancel

Name
AzureWebAppDemo20251010225432

Subscription name
Azure for Students

Resource group
AzureWebAppDemoRG (Canada Central) New...

Hosting Plan
AzureWebAppDemo20251010225432Plan* (Canada Central, S1) New...

☒ Use unique default hostname (preview).

Export... Create Cancel

Name
AzureWebAppDemo20251010225724

Subscription name
Azure for Students

Resource group
AzureWebAppDemoRG (Canada Central) New...

Hosting Plan
AzureWebAppDemo20251010225724Plan* (West US, F1) New...

☒ Use unique default hostname (preview).

Export... Create Cancel

Select existing or create a new Azure App Service

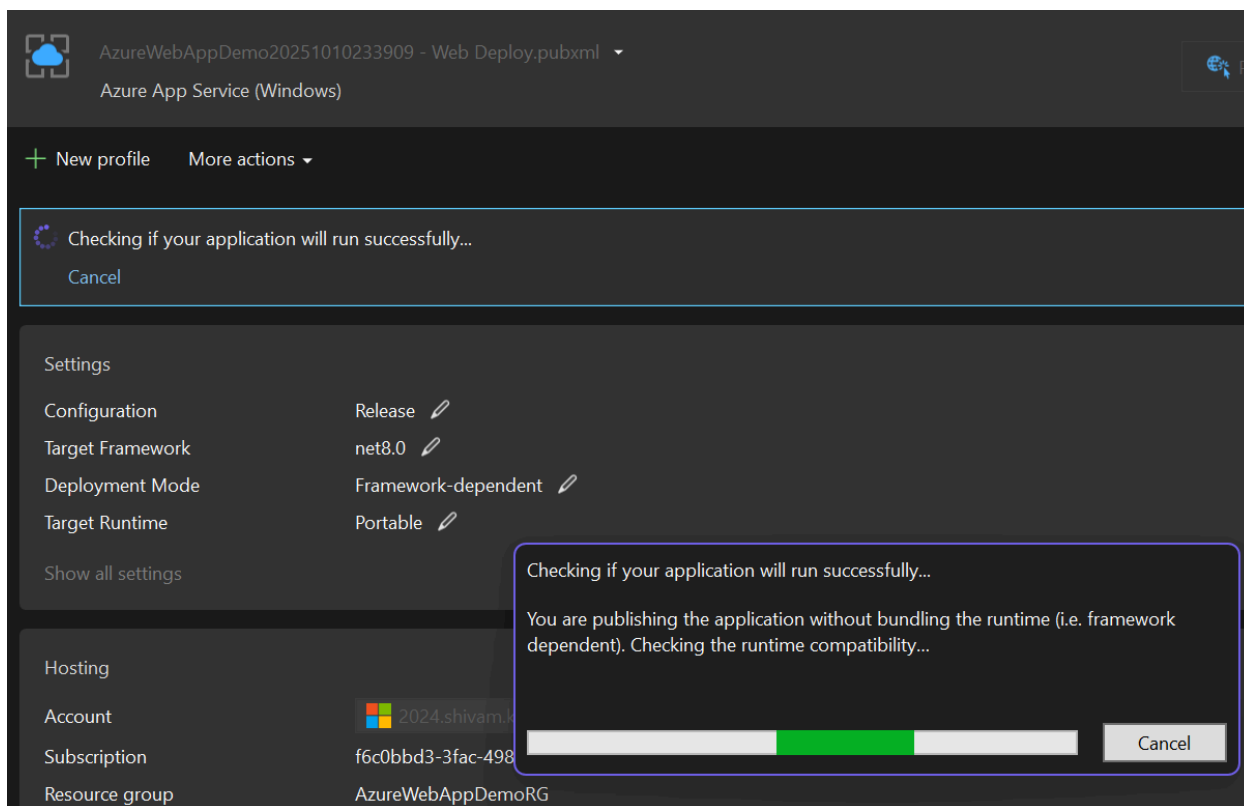
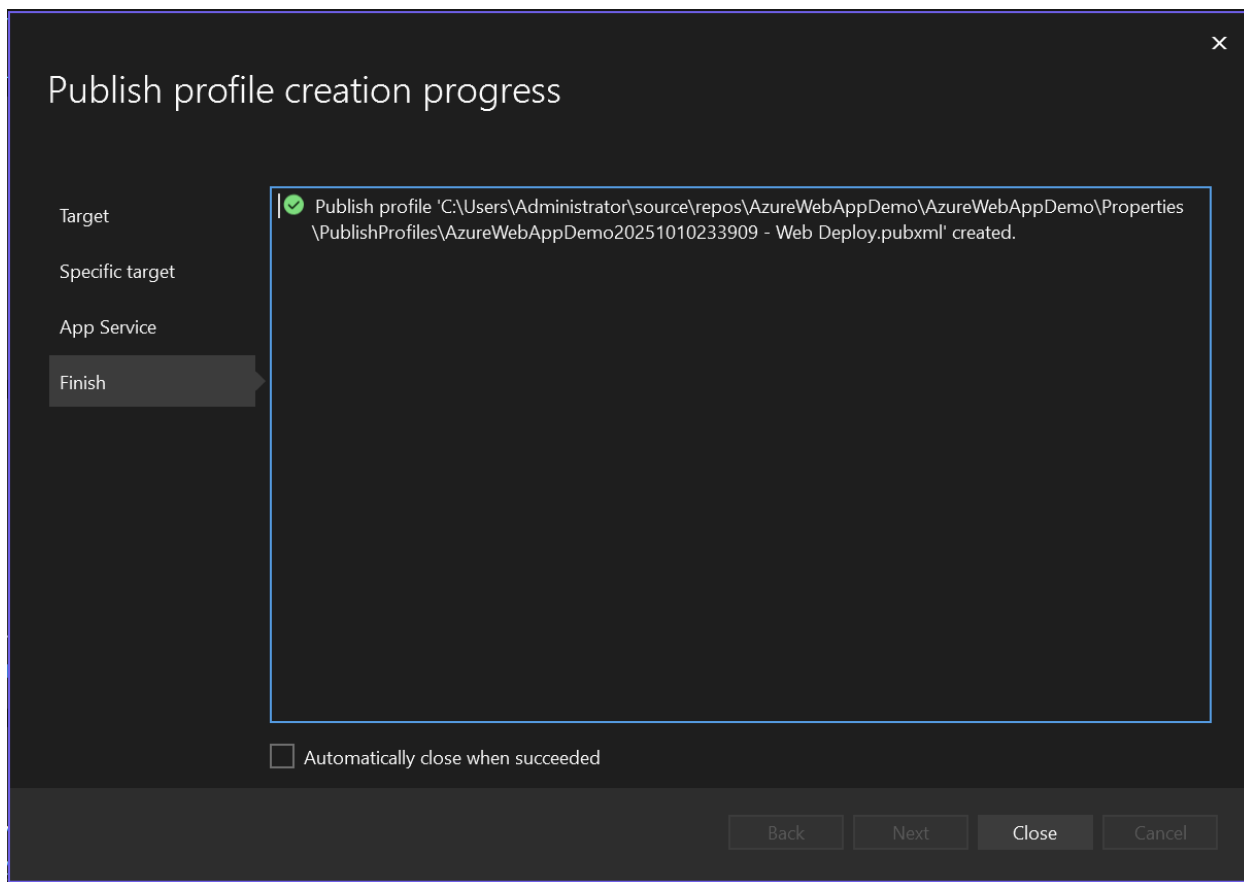
Target Subscription name
Azure for Students

Specific target
App Service AzureWebAppDemo20251010233909 + Create new [i] [r]

▲ AzureWebAppDemoRG
 ▲ AzureWebAppDemo20251010233909
 ▶ Deployment Slots

☐ Deploy as ZIP package
☐ Turn on Basic Authentication (not recommended) ⓘ

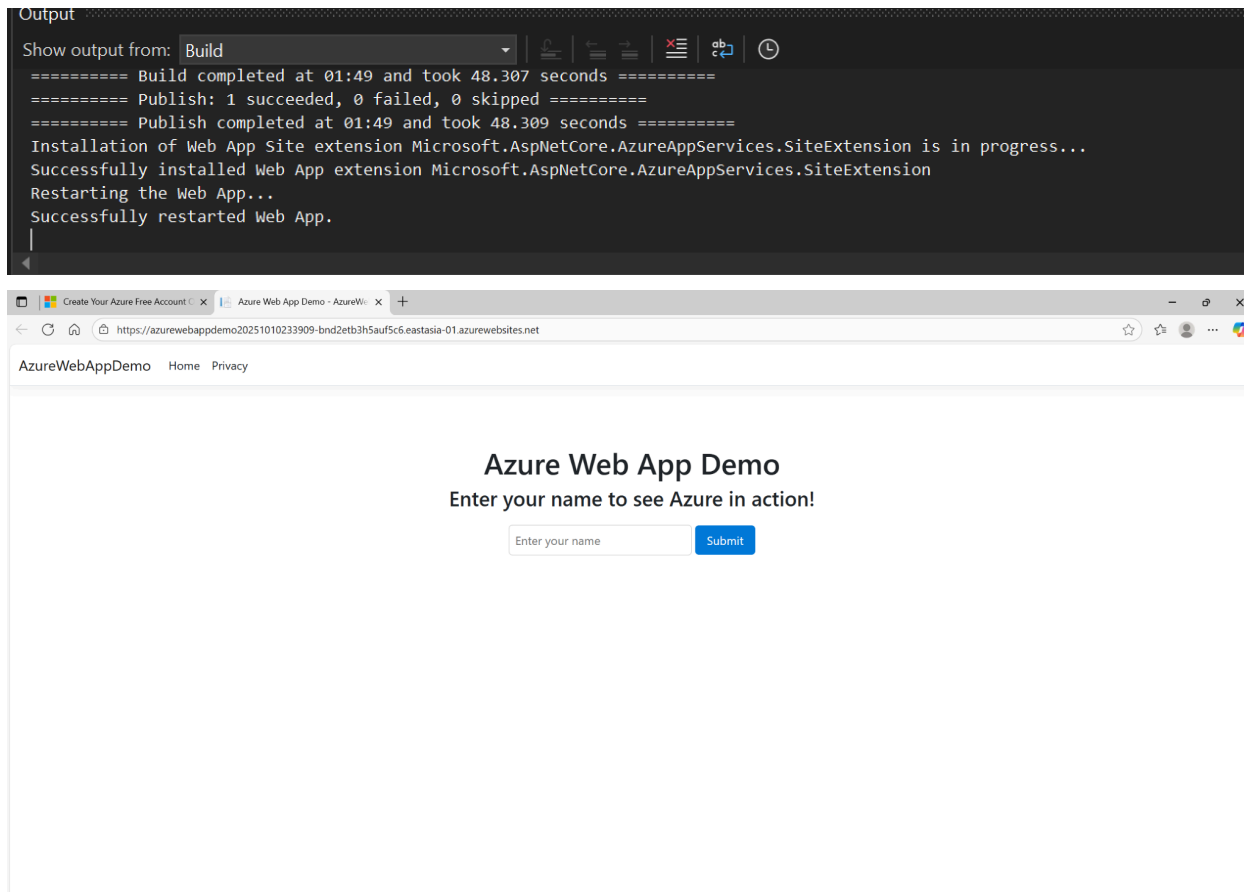
Back Next Finish Cancel

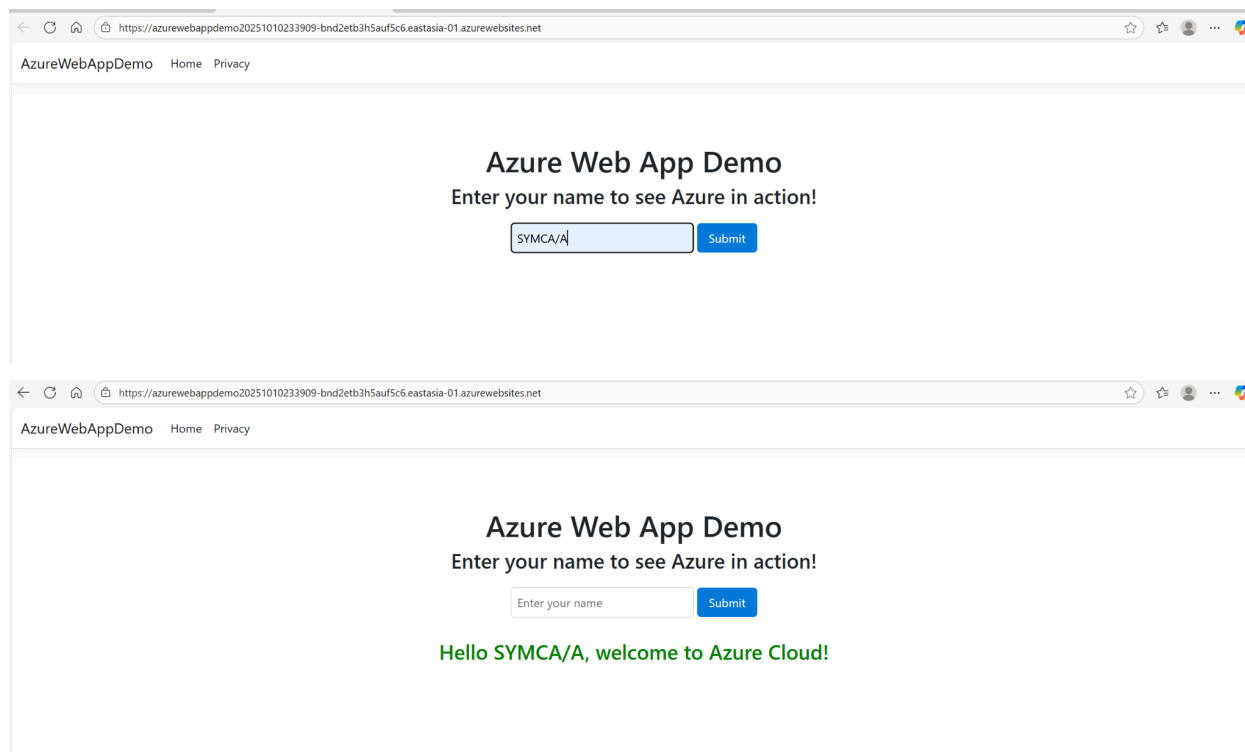


B. Verify

Deployment

- Once done, Visual Studio will open your deployed site automatically, e.g.:
<https://azurewebappdemo20251010233909-bnd2etb3h5auf5c6.eastasia-01.azurewebsites.net/>
- Enter your name again, and verify that it works as expected — hosted completely on Azure!





Conclusion :

The project successfully demonstrates the process of developing and deploying a web application using Microsoft Azure and Visual Studio. It highlights the advantages of cloud-based development such as scalability, reliability, and reduced infrastructure management. The use of ASP.NET Core MVC framework ensures modular, maintainable, and efficient application design.

By hosting the application on **Azure App Service**, the deployment becomes faster and easier compared to traditional on-premises hosting. This project also emphasizes how Azure's integration with Visual Studio simplifies cloud application lifecycle management. Overall, this exercise strengthens the understanding of cloud technologies, deployment automation, and the capabilities of Azure as a modern cloud computing platform.